PLANETARY COUNCIL · POLICY BRIEFING · OCTOBER 2026

Can Britain Feed Itself?

Store and grow: a plan for UK food resilience

Britain grows just enough calories in a normal year and holds no public food reserve. For about **£10 a person a year** it could store food for everyone, and for about **£4.50 more** it could start growing the soil, water, seed and nutrients that every future harvest depends on.

STORE
The UK Strategic Food Reserve
~£10.33 per person per year

GROW
Domestic resilience
~£4.46 per person per year of new money

Short edition, about 20 pages. The evidence edition (every claim, source and counter-argument) and both cost models are published openly at github.com/planetarycouncil/food-security-climate-regeneration. **Resilience through transparency: tell us where this breaks.**

At a glance

> The benefit first: safety and stability for about £10 a person a year.

The problem. Britain's farms grow about 101% of the calories the country needs, a margin of 31 kcal per person per day. That margin rests on imported gas-based fertiliser, imported soya for animal feed, a just-in-time supply chain and weather that is becoming less reliable. The government confirmed in March 2026 that the UK has "no national food stockpiles or plans to create these". Households are told to keep three days of food and water; the state keeps none.

The answer: two equally important pillars.

	STORE · UK Strategic Food Reserve	GROW · Domestic resilience
What it buys	Bread for everyone from 4 Mt of food-grade wheat; a 3-day no-cook ration and 90 days of vitamin tablets for all 69.5 million; formula and specialist food; next season's seed and safe nitrogen; ten weeks of soya meal; months of rice, oil and sugar	Soil testing and a soil-moisture network; farm and catchment reservoirs; UK seed multiplication; nutrient recycling; home-grown protein; support through the switch to regenerative farming
Protects against	Weeks to months: blockades, blackouts, cyber attacks, failed imports	Seasons to decades: fertiliser and feed dependence, drought, soil loss
Cost per person a year	**~£10.33** (£4.93 taxes, £5.40 food prices)	**~£4.46** new money, plus today's £4.87bn of farm support aimed at resilience
Together	**~£14.79 per person a year: under 5p a day**	

What it is worth. The plan insures a food sector that adds £162bn a year to the economy and employs 4.1 million people, for under 0.5% of that value. It pays for itself if it prevents one 1%-of-GDP shock in 40 years.

Above both pillars, a written contingency ladder: fair rationing, a livestock destocking order and safe nitrogen, agreed in advance and never improvised.

The decision worth taking now

1. Legislate a statutory food-reserve floor, published release triggers and a power to direct food businesses in a crisis.
2. Ask the ONS how compulsory commercial stocks would be classified, before any purchase.
3. Fund a business case and practical trials: a mill-by-mill grist trial, a local delivery drill with offline access and water, and a dietitian-designed ration with variants.
4. Start the soil-test and soil-moisture expansion, and scale the reservoir grant.

What we get: the value of safety and stability

> Safety is invisible when it works. That is why it is underpriced, and why it is worth paying for.

When a reserve is never used, nothing happens, and nothing happening looks like money wasted. Mark Carney called this the “tragedy of the horizon”: the cost of preparing arrives now, and the losses it prevents arrive later and land on someone else. Markets will value a space company in trillions on hope. Nobody prices the thing all those companies assume: that the people building them will eat this week.

So instead of promising a large benefit, we show **how small a benefit is enough** for the plan to pay its way.

Benefit	Pays for itself if... (store · both pillars)
Safety of the food system. A sector adding £162bn a year and employing 4.1 m people (Defra)	The premium is **0.44% · 0.63%** of the sector’s annual value added, about £175 · £250 per food job a year
Peace of mind. A plan, food and fuel for everyone	Average life satisfaction rises by **0.0008 · 0.0011 points** on a 0–10 scale (HM Treasury values one point for one year at £13,000). Too small to measure
Lives. Hunger, unsafe water, missed medicines and formula in a long disruption	It prevents **~290 · ~410 deaths a year** on average (DfT value of a prevented fatality, uprated)
A stable economy. Shops, mills and workplaces keep running	One week of 10% lower national output (≈ £5.6bn) costs as much as **~8 · ~5 years** of the programme
Social stability. Rising food prices and unfairness, more than empty fields, are followed by unrest	Not priced. In the 2000 fuel blockade, supermarkets were days from running out

Not priced, but real: food cannot easily be used as leverage against a country that holds stock; a buyer that does not have to panic-buy pays less in a crisis; seed and nitrogen in store protect next year’s harvest; the grow pillar rebuilds the soil every harvest stands on.

Scale. The UK spends about £62bn a year on defence, roughly £890 per person. The reserve costs about £10: around 1p of food security for every £1 of military security.

> These are break-even **thresholds**, not forecasts. Rows overlap and must not be added together. Values are in older price bases; updating them lowers the thresholds further.

CHAPTER 1

Can Britain feed itself?

> Resilient on paper, fragile in practice.

Britain produces about 60% of the food it eats by value, and on paper its farms produce just enough calories and nearly twice the protein people need. That paper picture hides three fractures. The protein depends on imported soya fed to animals. Yields depend on imported gas-based nitrogen. And the whole system runs on just-in-time logistics with no public reserve. Self-sufficiency, food security and resilience are different questions. This briefing is about resilience: what happens in the weeks and months after something breaks.

60%
of food eaten is produced in the UK, by value (2025); over 80% in the 1980s

31 kcal
per person per day: the margin between farm output and need (101%)

~1.1 Mt
of protein a year arrives as imported soya meal, mostly for pigs and poultry

0
national food stockpiles, and "no plans to create these" (Defra, Mar 2026)

Key facts

- By volume Britain is more than self-sufficient in milk, lamb, oats and barley, but grows only 83% of its wheat, 56% of its fresh vegetables and 14% of its fresh fruit (2025).
- Livestock use about 84% of farmland and supply about 31% of farm calories.
- Farm output covers ~101% of energy need *before* waste, and the release itself notes this would fall if imported feed stopped.
- 446,200 people work in agriculture; 40% of English farm holders are over 65 and 5% under 35.
- Food inflation peaked at 19.2% in March 2023.

Read with care. The 60% figure is by value and moves with import prices; it is not a calorie share. Defra's calorie statistic assumes no waste. Protein figures differ slightly within Defra's own release (193–195%).

Sources: Defra, *Agriculture in the UK 2025* (ch. 2, 10, 14); Defra, *Nutritional supply of agriculture in the UK 2025* (15 Oct 2026); written answer HL15829 (31 Mar 2026); Defra Farming Evidence Pack (Sep 2026).

CHAPTER 2

Nature’s gift

> World-class farmland per hectare, land-poor per person, with soils that can recover.

Few places grow cereals and grass as well as Britain: mild, wet, rarely frozen, rain-fed. The world wheat-yield record was set in Lincolnshire. But Britain has about a third of France’s arable land per person, and much of its best land sits in the drying east and on shrinking Fen peat. The lesson is not “grow more of everything”. It is to use the gift wisely: food for people on the best land, grass and trees on the rest, and soils rebuilt, as Britain’s own century-old experiments show is possible.

~7.8 t/ha
typical UK wheat yield, about twice the world average

17.95 t/ha
world record wheat yield, Lincolnshire, 2022

0.09 ha
of arable land per person, against ~0.28 ha in France

~£1.2bn
a year: the cost of soil degradation in England and Wales (2015 study)

Key facts

- About 71% of UK farmland is grass; England’s best and most versatile land is about 42% of farmland.
- 59.2% of England’s Grade 1 land lies in mapped river or sea flood-risk areas.
- Fen peat under arable loses 1–2 cm or more a year.
- At Rothamsted, a grass–clover ley rotation raised topsoil carbon from 0.98% to 1.23% over 28 years, then plateaued; continuous arable declined.
- Without action, England faces a public water shortfall of about 5 billion litres a day by 2055 (the Environment Agency’s do-nothing scenario).

Read with care. One record field is not the national system. Soil carbon gains are slow, plateau, and can be lost. A ley takes land out of crops for years. Mapped flood exposure is not the chance of loss in a given year.

Sources: Met Office; AHDB (2026); Guinness World Records; Defra AUK 2025 table 2.1; Environment Agency (2025, 2026); Graves et al. 2015, *Ecological Economics*; Johnston et al. 2017 (Rothamsted).

CHAPTER 3

Climate is changing the odds

> British wheat usually rides out a single bad spell. Bad weather stacked across a season is what hurts, and that is becoming more common.

British farming handles a single wet spell or heatwave. What breaks it is extremes stacked across one season: wet autumns that stop sowing, dry springs, wet harvests. In 2026 drought hit Britain, France and Germany together, so the neighbours Britain relies on struggled at the same time. New pests and livestock diseases are arriving with warmer winters. A reserve is the hedge against correlated failure that trade cannot provide.

4 of 5
of England's worst harvests since 1984 came in 2020–2026 (four crops, by volume)

−11%
2026 winter wheat yield against the five-year average (AHDB survey)

1,412
water-abstraction restrictions in England by late August 2026

1.2%
a year: chance all major wheat breadbaskets fail together (was 0.3%)

Key facts

- UK wheat losses come mainly from *combinations* of extremes within a season (Slater et al. 2022).
- 2026 was mixed: Defra's provisional England figures show wheat at 10.8 Mt (+0.8%), but barley down 11% and oats at their smallest since 2012. Full UK figures are due in December.
- France lost about 8 Mt of cereal output in 2026; EU soft-wheat yields fell about 7%.
- 2025 was the UK's warmest year on record. Wheat stem rust is re-emerging, and bluetongue is spreading.
- Wheat self-sufficiency fell from 96% (2023) to 79% (2024) and recovered to 83% (2025).

Read with care. Average yields may hold up for decades; the risk is in the bad years, not the mean. "Worst harvest" rankings depend on crops, geography and method. Some 2026 European losses reflect less planting because of input costs, not weather alone.

Sources: Slater et al. 2022, *Earth System Dynamics*; AHDB (2026); Defra provisional England estimates (8 Oct 2026); ECIU (Oct 2026); European Commission / JRC MARS; Met Office; Gaupp et al. 2020, *Nature Climate Change*.

CHAPTER 4

Hidden dependencies

> A loaf of bread runs on gas, diesel, ships and software.

Britain's farms turn imported energy into food. The country no longer makes ammonia at commercial scale. All its phosphate is imported, and much of it ends up in sewage. Food-grade CO_2, needed to stun animals and pack meat, comes from a handful of gas-dependent plants. Half of what we eat is chilled or frozen, and a ransomware attack on one distributor can stop orders to every major supermarket. Modern lorries need AdBlue (made from urea) as well as diesel, robotic warehouses need power and data, and wet grain needs fuel to dry. A week-long power cut would break the food system faster than any harvest failure.

2023
the year Britain's last large ammonia plant closed

~174 kt
of phosphorus imported a year; under half used productively

>50%
of the food we eat is chilled or frozen

20 km/h
the speed limit a Euro VI lorry is forced down to once its AdBlue runs out

Key facts

- Food-grade CO_2 has depended on a few plants; Billingham once supplied about 60%.
- In May 2025 ransomware at Peter Green Chilled halted orders to most major supermarkets.
- The Short Straits carry about 42% of dairy and egg imports and 39% of meat imports.
- The Iberian blackout of April 2025 stopped payments, warehouse systems and refrigeration for 50–60 million people.
- Food is one of 13 critical national infrastructure sectors, but the NAO says its actual assets (mills, dryers, CO_2, cold stores) still need to be identified.

Read with care. Phosphorus is a slow risk: soils hold legacy reserves. Manure recycles nutrients already in the system; it is not new supply. Attribution of the 2025 cyber attacks to one group is not established.

Sources: UK Food Security Report 2024; Lancaster UK Phosphorus Transformation Strategy (2022); Cold Chain Federation; Reg. (EU) 582/2011 Annex XIII; NAO, *Resilience of the food supply chain* (Sep 2026).

CHAPTER 5

Protein, livestock and the sea

> Grain isn't the weak point. Imported protein is. And our "animal" grain is a hidden reserve.

About half of Britain's wheat, barley and oats is fed to animals. Much of it could feed people if needed, as porridge, flatbreads and wholegrain, though not as today's sliced loaf without stronger wheat blended in. If imports stopped in a normal-harvest year, Britain's grain could keep people and animals going if industrial uses and exports were cut. The real gap is imported soya, which pig and poultry farming stands on. The answer: grow more protein at home, keep grass-fed cattle and sheep on land nothing else can use, size pig and poultry numbers to home-grown protein, and contract the winter mackerel and herring catch into cans and pouches.

~52%
of UK wheat, barley and oats goes to animal feed

4–6×
more calories go into grain-fed meat than come out

40%
of seafood eaten is now caught or farmed in the UK (89% in 1975)

What our grain model shows (imports stopped on 1 July; distilling and bioethanol cut by 75%; a 1 Mt working floor)

Shock over three years	Extra grain needed
Normal harvests, full livestock feed	2.3 Mt
Two harvests 15% down, feed at 80%	2.5 Mt
A nitrogen-shock harvest 35% down, feed at 80%	3.5 Mt
Three harvests 30% down, feed at 80%	11.4 Mt: storage alone cannot cover it; feed must fall to ~60% even with 4 Mt extra

Key facts

- Replacing imported soya protein fully would take ~4 Mt of field beans on ~1 Mha, about 16% of cropland; a third within a decade is a realistic aim.
- Only milk and upland suckler beef returned more human-edible protein than they consumed in a 2011 analysis of UK systems.

> **Read with care.** The model pools wheat, barley, oats and maize; a tonne of feed barley is not a loaf. It does not count the feed and CO_2 lost when distilling stops. Cutting UK livestock while importing meat moves emissions abroad.

Sources: Defra; AHDB; Mottet et al. 2017, *Global Food Security*; Wilkinson 2011, *Animal*; Harrison et al. 2023; MMO 2024; our cereal model (`cereal_balance.py`).

CHAPTER 6

Geopolitics and prices

> Two chokepoints in one year. Britain buys from neither directly, and pays for both.

In 2026 the Strait of Hormuz closed and Black Sea grain shipping came under attack. Fertiliser prices doubled within weeks and world food prices hit their highest level since 2022. In every big crisis, exporting countries ban exports, and the supply that remains flows to whoever holds stock. Britain has felt little so far, because the shock arrives with a lag through next year's fertiliser and harvest. A reserve is foreign policy: it stops Britain panic-buying, lets it keep its own exports open and help partners, and makes it a reliable ally rather than a competitor for scarce supplies.

up to 30%
of traded fertiliser passes through Hormuz

$400 → $850
per tonne of urea, early to April 2026

136.0
FAO Food Price Index, Sep 2026: highest since Nov 2022

77
food export restrictions by 32 countries in 2022

Key facts

- At the 2022 peak about 17% of traded food and feed calories were under export restrictions.
- UK food inflation was 1.3% in September 2026; spring warnings of 9% have not yet materialised.
- The UK's carbon border tax starts on imported fertiliser in January 2027.
- WTO rules allow temporary export restrictions in a food crisis, but exempt World Food Programme humanitarian purchases.
- The ASEAN+3 rice reserve (787,000 t earmarked) shows neighbours can share stock in calm markets.

Earning without gouging. In calm years a reliable reserve can earn modest income from storage, options and rotation sales to aid buyers, at prices agreed in advance, never in a crisis. The case is cost-sharing and influence, not profit.

> **Read with care.** Options and swaps fail when export bans bite, so a protected domestic floor comes first. Whether Hormuz is open in late October 2026 is unconfirmed.

Sources: FAO (2026); WTO data blog (Jul 2026); IFPRI export-restriction tracker; ONS (Sep 2026); HM Treasury CBAM factsheet; ASEAN APTERR.

CHAPTER 7

Food and stability

> Rising prices, more often than empty fields, are what break the peace. People forgive scarcity sooner than unfairness.

“There are only nine meals between mankind and anarchy,” wrote Alfred Henry Lewis in 1896. The modern evidence is more precise: rising food prices, not price swings, are followed by unrest. In rich countries, logistics fail first. In the 2000 fuel blockade, supermarkets warned they would run out within days. What turns shortage into disorder is unfairness: governments cutting bread subsidies overnight, ministers who “never buy rice”, stockpiling by those who can afford it. Fair, visible preparation is the stability policy.

12.2%
of UK households are food-insecure (Jul 2026)

60%
of households have three days of no-cook food; **4%** have three days of water

2.6 m
emergency food parcels handed out in 2025

Key facts

- Rising food price *levels* are followed by more social unrest across countries; volatility is not (Bellemare 2015).
- Egypt, 1977: cutting the bread subsidy triggered riots in nearly every major city; it was reversed within two days.
- Japan, 2025: rice prices nearly doubled, released reserve rice reached shops slowly, and the farm minister resigned.
- WWII rationing gave everyone the same basic entitlement; infant mortality fell during the war.
- The NAO (September 2026) found local food-disruption planning “insufficient”, most resilience forums without a food plan, and no power for Defra to direct food businesses in a catastrophe.
- In October 2026, over 150 experts wrote to the Prime Minister calling for national food stores.

Read with care. In Britain the realistic failure is panic buying, empty shelves and lost trust, not mass riots. Unrest has many causes; prices are one.

Sources: Quote Investigator; Bellemare 2015, *AJAE*; Food Foundation (2026); UK public preparedness survey (2026); Trussell; NAO (Sep 2026); The Grocer (5 Oct 2026).

CHAPTER 8

How other countries prepare

> Several comparable countries keep food in reserve. Britain keeps none.

Countries similar to Britain treat food as critical infrastructure. Finland must hold at least six months of grain. Switzerland makes importers hold two to four months of key foods, feed and fertiliser, and drew on its feed reserve this summer when drought stalled river transport. Norway is rebuilding three months of bread grain; Sweden's stockpile law took effect in July 2026; Germany is adding ready-to-eat meals for power cuts. Japan shows what goes wrong when a reserve is run down and refilled too late.

Country	What it holds	How
Finland	At least 6 months of bread grain (about 9 after a 2022–23 top-up)	State agency, funded by a supply-security fee
Switzerland	2–4 months of wheat, sugar, rice, oils, coffee, feed and fertiliser	Compulsory private stocks, rotated through firms' trade
Norway	Rebuilding ~82,500 t of bread grain (~3 months) by 2029	State-owned grain, millers store it on long contracts
Sweden	Grain, seed, fertiliser, feed and spare parts	New law from 1 July 2026; private custodians; sites confidential
Germany	Federal grain reserve (~1 month of bread) plus rice and pulses; ready meals planned	State reserve in ~150 private stores
Japan	Rice reserve, target 1 Mt	State buys, holds ~5 years, sells old rice as feed
Singapore	Two months of rice imports	Licence condition on importers
UK	**None**	

Lessons we take

- Compulsory commercial stocks work, and rotate through normal trade (Switzerland, Singapore).
- In August 2026 Switzerland announced the release of up to 16,000 t of protein feed because low Rhine and Danube water stalled imports.
- Japan's 2025 release was slow and late: rules for release and refill matter as much as tonnes.
- India shows the cost of poor storage: wheat worth ₹700 crore deteriorated in Punjab (2011–16).

Read with care. Finland's fee and Switzerland's cost (~CHF 14 per person, 2019) cover fuel and medicines too, so they are not like-for-like prices for a UK food reserve. Most national reserve data is incomplete or secret.

Sources: NESA (Finland); réservesuisse (1 Jan 2026); Norwegian government; Riksdag / SFS 2026:837; top agrar; Jiji / Asia News Network; MTI Singapore; CAG India.

STORE: the UK Strategic Food Reserve

> About £10 per person per year, under 3p a day, buys bread and a no-cook ration for everyone, the seed and nitrogen for the next harvest, and months of imported staples. Human food comes first, and nobody is left out.

A grain silo is not yet a meal, so the reserve is built backwards from the meal, and it is built for **everyone**.

> *Think of it as the National Insurance (not National Insurance).*
>
> Prof. Gen. Adm. F.L. Amingo, Battalion of Militarized Satire

Bread first. 4 Mt of food-grade wheat, held by millers and merchants on top of their normal stock, equals about ten months of today's milling with no imports and no harvest, as long as the mills have power, people and the right grist. A strong-wheat tranche keeps the loaf rising; a premix keeps flour fortified. Grain keeps for years when dry and monitored. Each year about a fifth rotates into mills, animal feed, or abroad as export or aid.

No-cook for everyone. When the power goes, cookers and bakeries stop. So every person is covered by a **3-day no-cook grain ration** (crispbread, biscuits, oatcakes from reserve grain), stored locally and rotated into schools, the forces and food banks. Vitamin tablets cover everyone for 90 days. On top come **specialist needs**: infant formula, medical foods, and soft or ready meals for people who cannot use the standard ration.

The next harvest and the rest of the diet. Compulsory stocks hold winter and spring seed, extra pre-season nitrogen held safely as urea and UAN (never solid ammonium nitrate, the Beirut chemical), ten weeks of soya meal, and months of rice, oil and sugar. The state pays for priority fuel, AdBlue and backup power at mills, depots and dryers, and for local plans in all four nations.

Two tiers. A **public crisis tranche** owned by the state and paid from taxes, and **compulsory commercial stocks** (the Swiss model) that firms hold on top of their normal stock and rotate through their own trade, with the cost passed through food prices.

If money is short, cut from the bottom (cumulative per person a year):

Up to	Buys
£2.71	**Bread for everyone:** 4 Mt food-grade wheat, strong wheat, premix, sacks
£3.48	+ **no-cook food for everyone:** 3-day ration, tablets, bottled water
£6.92	+ operations: agency, local plans, priority fuel and power, last mile
£7.55	+ infants and special needs
£9.19	+ next season: seed, safe nitrogen, soya meal
£10.33	+ imported staples: rice, oil, sugar, pulses, oats, milk powder, fish

The reserve in numbers

Component	Stock	£ per person a year
Food-grade milling wheat 4 Mt (half in new stores) + 0.35 Mt strong wheat	4.35 Mt	2.62
No-cook grain ration for everyone, 3 days (~500 g a day)	104 kt	0.40
Tablets, flour premix, sacks, 90 m litres of bottled water	–	0.46
Operations: lean agency and inspectorate (~£33m), local plans, priority fuel and power, last mile, security	–	3.44
Infant formula (~7 kt), medical foods, specialist no-cook top-up for 5 m people	–	0.63
Next season: cereal seed, protected urea and UAN-28 (~162 kt N), 10 weeks of soya meal	1.21 Mt	1.64
Imported staples: rice and oil (~4 months), sugar (~2 months), pulses, oats, milk powder, canned fish	1.18 Mt	1.14
Total	**~6.8 Mt**	**~£10.33 (£718m)**

£4.93
via taxes (public tranche)

£5.40
via food prices (compulsory stocks)

~£3.7bn
of stock and stores to finance up front, public and private, over ~5 years

Range (sensitivities, not a confidence interval): low £9.40 · central **£10.33** · prices +30% £11.35 · firms financing at 8% £11.92, at 10% £12.73 · grain and fertiliser prices doubled £13.73 · stress £16.56 · severe £18.34.

What could go wrong, and how it is handled

Risk	Answer
It adds to public debt	Only the ~£0.72bn public tranche is designed to sit on the balance sheet. If the ONS classifies the compulsory stocks to government, debt rises by ~£2.2bn more. Ask first
Firms relabel normal stock	Obligations on top of a registered baseline, checked by an £8m-a-year independent inspectorate
Week one fails without water and delivery	90 m litres covers 10 m people for 3 days; water companies' duties cover the rest and must be drilled. One national food drop would cost ~£400m extra to deliver
The same wheat is counted twice	The 4 Mt is the bread floor only; keeping animals fed through two poor harvests needs ~2.5 Mt of separate feed grain
Stored fertiliser explodes	No solid ammonium nitrate: protected urea and heated UAN-28 only
Refill forces bad buying	Refill on a price-and-quality trigger within 36 months; refilling 4 Mt 30% dearer costs ~£0.25bn
It is raided to fight inflation	Release only on published supply triggers, never to manage prices

GROW: domestic resilience

> A reserve buys time. Soil, water, seed and home-made nutrients decide what we can grow with it.

Britain's farms grow about as many calories as the country needs, but every harvest rests on imports and luck. The nitrogen comes from gas made into ammonia abroad. The protein for pigs and poultry comes from soya across the ocean. The phosphate is imported, and much of it ends up in sewage. The rain is getting less reliable. A store of food covers a bad season; it cannot cover a decade of dependence. So the second pillar grows resilience at home, and it is just as important.

Component	What it buys	£m a year
Soil and water monitoring	~500 soil-moisture sites (from ~50), satellite moisture, a sampled national soil-health baseline, then regular field tests	35
Water storage	Farm and catchment reservoirs that fill in winter: roughly 17,000–50,000 megalitres of new storage a year	100
Seed security	Gene banks, UK seed multiplication (incl. spring cereals and pulses), public breeding	40
Nutrient independence	Manure and digestate testing and precise use; phosphorus recovery; legumes; a green-ammonia pilot	60
Regenerative transition	Independent advice, farmer networks, bridging payments, ~12,000–29,000 ha a year	50
Home-grown protein	Dehulling, pulse mills, local processing	25
Total new money	**~£4.46 per person a year**	**~310**

Existing money, better aimed. The UK already spends about £4.87bn a year on farm support, roughly £70 per person. The grow pillar asks that this money buy resilience: soil health, water, protein crops, nutrient efficiency, and food production on the best land. It does not ask for more money, but retargeting has losers as well as winners, and they should be named.

Key facts

- England targets 40% of farm soils under sustainable management by 2028 and 60% by 2030 (management actions, not measured recovery).
- About 50 soil-moisture stations cover the whole UK today.
- Reservoir grant rounds have been £10–15m; the August 2026 round was £15m for about 100 farms.
- The UK Vegetable Genebank holds about 14,000 seed samples.

> **Read with care.** £310m a year is a starting envelope and a set of pilots, not national nutrient independence. Regenerative farming often improves annual profit after 3–5 years, but paying back the investment takes nearer nine years without support. Pay for measured outcomes, not labels.

Why it matters beyond food



The 16 benefits, checked against the evidence. Supported: healthier soils at specific sites, erosion control, biodiversity on hedges, margins and flower-rich leys, wellbeing from structured outdoor work. Partly supported or conditional: water cycles, lower input costs, climate resilience, cleaner air (only if fertiliser rates fall and slurry is covered), local jobs (more labour per hectare, but net jobs not shown). Weak or mixed: nutrient density. Real but limited: soil carbon rises, then plateaus. Values, not measurements: education, community, spiritual connection, future generations. The evidence edition sets out each label and its source.

CHAPTER 11

Running it: a lean, open, automated agency

> About 65 people, three deep, heavy on automation, open by default. You cannot hide a thousand jobs and forty warehouses, so we don't try.

The reserve is run by a **lean agency of about 65 people**, never more than 100 even in a crisis. Every function has a lead and two trained deputies, so leave, illness or resignation never stops anything. Automation does the routine work: sensors in every store, automatic samplers and analysers at the hubs, and AI watching stock, prices, weather and shipping around the clock. People decide, check, sign and answer to Parliament. Physical inspection is contracted to an independent inspectorate. A small innovation team keeps improving nutrition, storage, transport, sampling and data.

The network: near ports, near people

- **~8–10 port hubs** at bulk-grain ports such as Immingham, Tilbury, Liverpool, Avonmouth, Ipswich and Belfast (not Dover, which is a lorry port).
- **~30 district stores** across all four nations, much of the stock inland, away from the Humber and Wash flood zones.
- **Thousands of local points** (schools, councils, depots) holding the no-cook ration.
- A **statutory minimum in Northern Ireland**.

Open by default. The agency publishes its host towns and ports, monthly regional stocks, buying rules and quarterly volumes, costs, audits and incidents. A short, public list of exceptions is withheld: exact fuel, formula and nitrogen sites, live release routes, and precise purchase dates. It runs a live dashboard, a citizens' panel, an annual public stress test and a bounty for anyone who finds a real weakness.

~£33m
a year for the agency, sensors, data and inspectorate

2 people
must sign every release of public stock

~1,070
gross jobs across stores, labs, haulage and local planning

Read with care. Sixty-five is a commissioning team, to be resized from real workloads; the 24/7 room alone takes about 11 posts. Sensors screen for problems; accredited labs still certify food. Cloud AI can fail first in a blackout, so everything must work on paper and radio, and drills run without the AI.

CHAPTER 12

Work, prestige and a food-producing nation

> Make farming a career people are proud of, let machines do the drudgery, and grow a surplus Britain can sell in good years and eat in bad ones.

Two things are happening at once. AI and automation are hollowing out entry-level office work: graduate adverts are at their lowest since 2016 and youth unemployment is above 16%. Meanwhile Britain's farmers are ageing. Both pillars need people: in stores, labs and local planning, and as soil testers, water engineers, seed breeders, nutrient advisers and pulse processors. Good work outdoors, with others and with purpose, measurably improves wellbeing. But farm work today is often isolated, seasonal and stressful, so the jobs must be redesigned, not just advertised: decent pay, training ladders, housing, teams, and machines for the repetitive tasks.

A food-producing nation. Britain will never grow everything it eats. But it can produce more of what it grows best and sell the surplus: oats and barley, dairy and milk powder, lamb, seed potatoes, seed and breeding expertise, and premium, high-standard food. A surplus works as a reserve only for what is already in surplus, and only if it can be kept at home in a crisis.

up to ~3,600
gross jobs across both pillars (not net of displaced work)

£25.6bn
of food and drink exports in 2025, against £66.9bn of imports

−0.64 / −0.94
effect of structured outdoor activity on depression / anxiety

Key facts

- In the 2020 Pick for Britain campaign, UK residents filled only about 5–11% of picking roles: schemes alone do not move people onto farms.
- The US Civilian Conservation Corps (1930s) raised participants' health, longevity and lifetime earnings.
- 91% of young farmers call poor mental health the industry's biggest hidden problem.
- A work-in-nature pilot (10,000 paid places, ~£350m a year) would be a separate, separately funded decision.

Read with care. No study yet shows people displaced by AI moving into land work. Prestige follows pay, land access and housing, not slogans.

Sources: Adzuna; ONS; Coventry et al. 2021; NBER w27103; NFU (2020); Farm Safety Foundation; FDF trade snapshot 2025.

CHAPTER 13

A new economy of food

> Price what harms, pay for what protects, and give the money back.

Cheap food has hidden costs: emissions, polluted rivers, degraded soils and diet-related disease. Carbon pricing is reaching farming anyway, starting with a border tax on imported fertiliser in January 2027. Carbon money is shifting towards durable removal while soil-carbon offsets remain unreliable. Denmark shows a workable model: tax livestock emissions, deduct most of it, and recycle the revenue into farming. The lesson of Europe's 2024 farmer protests is that change imposed on farmers fails, and change that pays them works.

Our position. One consistent carbon price across jurisdictions: no free allowances, no subsidies for emitting, and CO_2 treated as a commodity. Emitters pay; verified removers earn. In practice: link the UK and EU schemes and phase out free allowances jointly, as linking requires; price avoided emissions, temporary storage and durable removals as different products; phase out fossil breaks such as the red-diesel rebate gradually, returning the money to the same farms; and pay for verified, durable removals as a product.

~12%
of UK greenhouse gas emissions come from agriculture, including 49% of methane and 59% of nitrous oxide (2024)

≥ $10 trillion
a year: the hidden costs of the world's food system (FAO)

£0.4–1bn
a year: Danish-style livestock tax rates applied to UK livestock, all recycled (our estimate)

Key facts

- UK agricultural support was £4.87bn in 2025, including a red-diesel rebate of about £384m.
- Soil-carbon credits have been shown to track rainfall more than management in Australia.
- Biodiversity net gain statutory credits start at £42,000 each, with two needed per unit: a deliberately expensive last resort.

Read with care. Full carbon pricing raises food prices, so a per-person dividend matters. Pricing UK livestock without a border adjustment on food imports exports the emissions. Do not spend the same revenue twice.

Sources: Defra AUK 2025 ch. 10–11; FAO SOFA 2023; Danish green tripartite agreement; Mitchell et al. 2024; GOV.UK statutory biodiversity credit prices; UK ETS Authority (Nov 2025).

CHAPTER 14

The plan

> Store and grow. Plan the next step. Fewer, better animals. Close the loops. Price the truth. Bring people with us.

When	Store	Grow
0–12 months	Legislate the reserve, a statutory minimum, release triggers and a power to direct; ask the ONS; buy food-grade grain only in good harvests; local food plans in every resilience forum	Start the soil-test and soil-moisture expansion; scale the reservoir grant; aim farm support at soil, water, protein and nutrient efficiency
1–3 years	Phase in compulsory stocks; build the public tranche and the universal no-cook ration; agree and exercise the destocking order and rationing rules	Seed multiplication and public breeding; first reservoirs filling; manure testing; phosphorus recovery; pulse processing
3–10 years	Mature, rotating, audited reserve in all four nations	Soya imports down by about a third; national soil-health map; green-ammonia pilot

The contingency ladder

Level	Situation	What happens
0 · Normal	Business as usual	The reserve rotates; monthly public stock report; exercises
1 · Disruption	Days to weeks: fuel blockade, cyber attack, blackout	No-cook food, priority fuel and power for mills and depots, purchase limits
2 · Sustained shortage	Months: import cut, input shock, one failed harvest	Staged release; feed-to-food switching; distilling cut; destocking stage 1
3 · Severe	Two poor harvests plus fertiliser shortage	**Fair Shares Guarantee**: the same basic entitlement for everyone, top-ups by need, capped staple prices, paper fallback, automatic exit
4 · Catastrophic	Volcanic winter, war, global crop disease	All edible grain to people; international mutual aid

The livestock destocking order, written in advance with farmers and vets: stop new births first; simplify processing to bulk cuts for public catering; keep lawful by-product routes open; pre-agreed compensation; and never let animals starve.

The law. Emergency regulations under the Civil Contingencies Act must be approved within 7 days and expire after 30. A Food Security (Emergency) Act should build the standing scheme; the emergency powers switch it on. The **Cabinet Office holds the release key; Defra operates**, with all four governments at the table.

RELATED CONCEPT · NOT PART OF THE COSTED PLAN

Homes, Land and Livelihoods

> Build homes where the work is. New settlements can give a new generation a home, land to work and a reason to stay.

Britain has built new places before when it needed them: the garden cities of Letchworth (1903) and Welwyn (1920), the post-war new towns, Milton Keynes (1967), Cambourne in Cambridgeshire (from 1998), and now Salt Cross garden village near Eynsham, whose plan for about 2,200 homes and a science park was adopted in February 2026. The government's new-towns programme is choosing the next sites. Since COVID, many people no longer have to live near an office.

That opens a different kind of settlement: modern, efficient homes **clustered** near an existing village or town, with **shared** farmland held by a community land trust, a machinery ring, a pack house and a cold store, so residents can work in food, nature and land management for some or all of the week. The modern "mule" is shared kit, skills and working capital, not a plot each.

Design rules

- **The home never depends on the farm job.** If a farm business fails, residents keep their homes.
- **Affordability comes from tenure**, through a land trust or housing association, not from a "smallholding" label.
- **Keep the best land in food**: build on the edge of existing places or on previously developed land, never on peat or flood plain.
- **Count whole-life carbon**, including car journeys; rural is not automatically low-carbon.
- **Pilot two or three sites first** against ordinary affordable housing plus a separate farm-entry scheme.

1.62 ha
four acres per household would take ~9.6% of UK farmland for a million homes

~£500k
of sales a shared farm needs to pay four people; the village alone buys ~£52k

Read with care. More homes help affordability locally, but national prices are driven mostly by interest rates (Miles & Monro, Bank of England, 2019). Planning is slow and contested. The phrase "forty acres and a mule" is not used: it was a broken promise of land to formerly enslaved Americans.

Resilience through transparency

> We publish our own critique. Tell us where this breaks.

This briefing has been through four rounds of independent adversarial review. Every claim in the evidence edition carries a label: **[P]** primary source, **[S]** secondary source, **[E]** our estimate, **[I]** interpretation or position, **[?]** not yet verified.

What still needs checking before anyone spends money

1. How the ONS would classify compulsory commercial stocks, and what firms would really pay to finance them.
2. Whether mills can run emergency grists, plant by plant, with backup power.
3. A dietitian-designed ration with variants for infants, pregnancy, coeliac disease, allergies and people who cannot chew.
4. A priced last-mile and water service, tested in a real local drill.
5. Quotes for storage, new stores, reservoirs and fuel contracts.
6. Full UK 2026 harvest figures (due December 2026) and Defra’s reply to our FOI request.

Open materials

- Evidence edition: every claim, source and counter-argument.
- Two Python models: the cost of both pillars (`reserve_basket.py`) and a monthly grain-stock model (`cereal_balance.py`).
- 27 research notes.

github.com/planetarycouncil/food-security-climate-regeneration

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An army defends the border. A food reserve defends the dinner table.